SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I S.A. Balaji/192126014 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: BJ

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	1
Explanation	25%	1	1	1	1
Application	20%	1	1	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 27 **/40**

Signature of the Faculty

Computer Network

27

CO-1 AT-1

Name: S.A. Balaji (7)
Reg No: 192126011L6
Course: Computer Network
Code: CSM0709
Course Faculty: Dr. R. Shankar

1. A company need to transfer a 150MB design file from a client computer to a remote server over a network with bandwidth of 50Mbps. The transport layer divide the file into 1500-byte segments, while the Data link layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by Data link layer, and estimate the total transmission time. Explain how the transport and Data links layers ensure the reliable Communication.

2. A transport layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If sender transmits 20 frames. calculate the number of transmission window required. If one frame in every window is lost and must be transmitted. determine the total retransmission. Explain how flow control improves communication efficiency.

3. A 12,000 byte IP packet is transmitted through a router where the maximum transmission unit (MTU) is 1500 bytes assuming each fragment requires a 20-byte IP header. Calculate the total number of fragments created and the header overhead introduced. Explain how network layer ensures successful delivery of fragmented packets to destination

A Transport layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If sender transmits 18 frames, calculate the number of transmission window required. If one frames in every window is lost and must be retransmitted, determine the total number of retransmission. Explain how flow control improves communication efficiency.

1. FILE TRANSFER, SEGMENTS, FRAMES, AND TRANSMISSION TIME.

Given:

→ File size = 150MB

→ Bandwidth = 50Mbps

→ Segment size = 1500 bytes.

→ Data link layer overhead = 40 bytes per frame.

Number of segments:

→ $\frac{150 \times 10^6}{1500} = 100,000$

no. of segment = 100,000

Number of frames:

→ Assuming one segment is carried by one frame.

→ number of frames = 100,000

Total Data link layer Overhead:

100,000 x 40 = 4,000,000 bytes.

Overhead = 4MB.

Total transmission time:

Total data transmitted:

→ 150 + 4 = 154MB

$T = \frac{154 \times 8}{50} = 24.64$ seconds.

Transmission time = 24.64 seconds.

How reliable communication is ensured.

→ The transport layer uses acknowledgements, sequence numbers, and retransmissions to ensure a lost data.

→ The Data Link layer uses frame checking such as CRC, and may retransmit damaged frames.

2. Sliding Window and Retransmission given:

→ sliding window size = 6 frames
→ Total frames = 48.
→ One frame is lost in every window

1. Number of transmission windows:

$$\text{windows} = \frac{\text{Total frames}}{\text{window size}}$$

$$= \frac{48}{6}$$

windows = 8

2. Total number of retransmission.

One frame is lost in each window.

Therefore,

Retransmission = 8 × 1

= 8 frames.

3. How does flow control improve communication efficiency.

→ Flow control regulates the rate at which the sender transmits data to the receiver. It prevents the receiver's buffer from overflowing and reduces data loss.

→ The sliding window allows multiple frames to be sent before receiving acknowledgement, improving network utilization and communication efficiency.

3. IP Packet fragmentation

Given:

-> IP packet size = 12,000 bytes.

-> MTU = 1500 bytes.

-> IP header per fragment = 20 bytes.

1. Data carried per full fragment:

-> 1500 - 20 = 1480 bytes.

Assuming the original 12,000 byte IP packet includes its 20-byte header.

Original payload = 12000 - 20 = 1.

fragments = $\left\lceil \frac{11980}{1480} \right\rceil = 9.$

2. Total header overhead:

$9 \times 20 = 190$ bytes.

Answer:

-> 9 fragments and 190 bytes of fragment-header overhead.

3. How the Network layer ensures delivery.

-> Divides large packets into a smaller fragments when necessary.

-> Uses identification and fragments offset fields to supports reassembly.

-> The destination reassembles the fragments into the original packet.

-> Error detection and higher-layer retransmission can help recover the lost data.

5. Session layer Synchronization:

given:

-> Total file size = 600MB

-> Checkpoint interval = 60MB

-> Failure occurs after 390MB.

1. Number of checkpoints

$$\frac{390}{60} = 6.5$$

completed checkpoints occur at 60, 120, 180, 240, 300 and 360mb

2. Data to be retransmitted:

390 - 360 = 30MB.

→ 6 checkpoints and 30MB must be retransmitted, assuming recovery resumes from the latest completed checkpoint.

3. Importance of synchronization:

→ Creates a recovery points during transmission.

→ Prevents restarting the entire transfer after a failure.